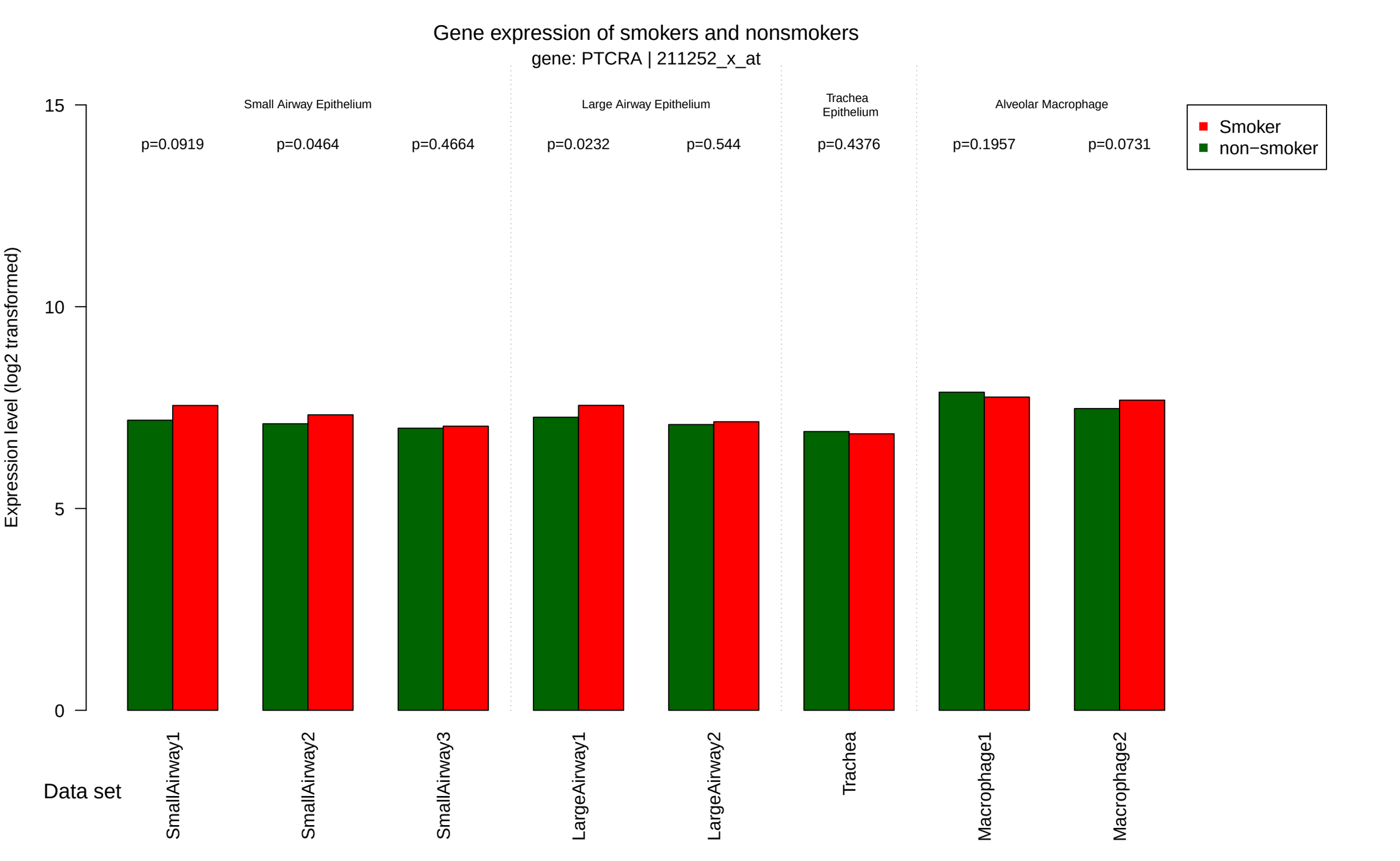
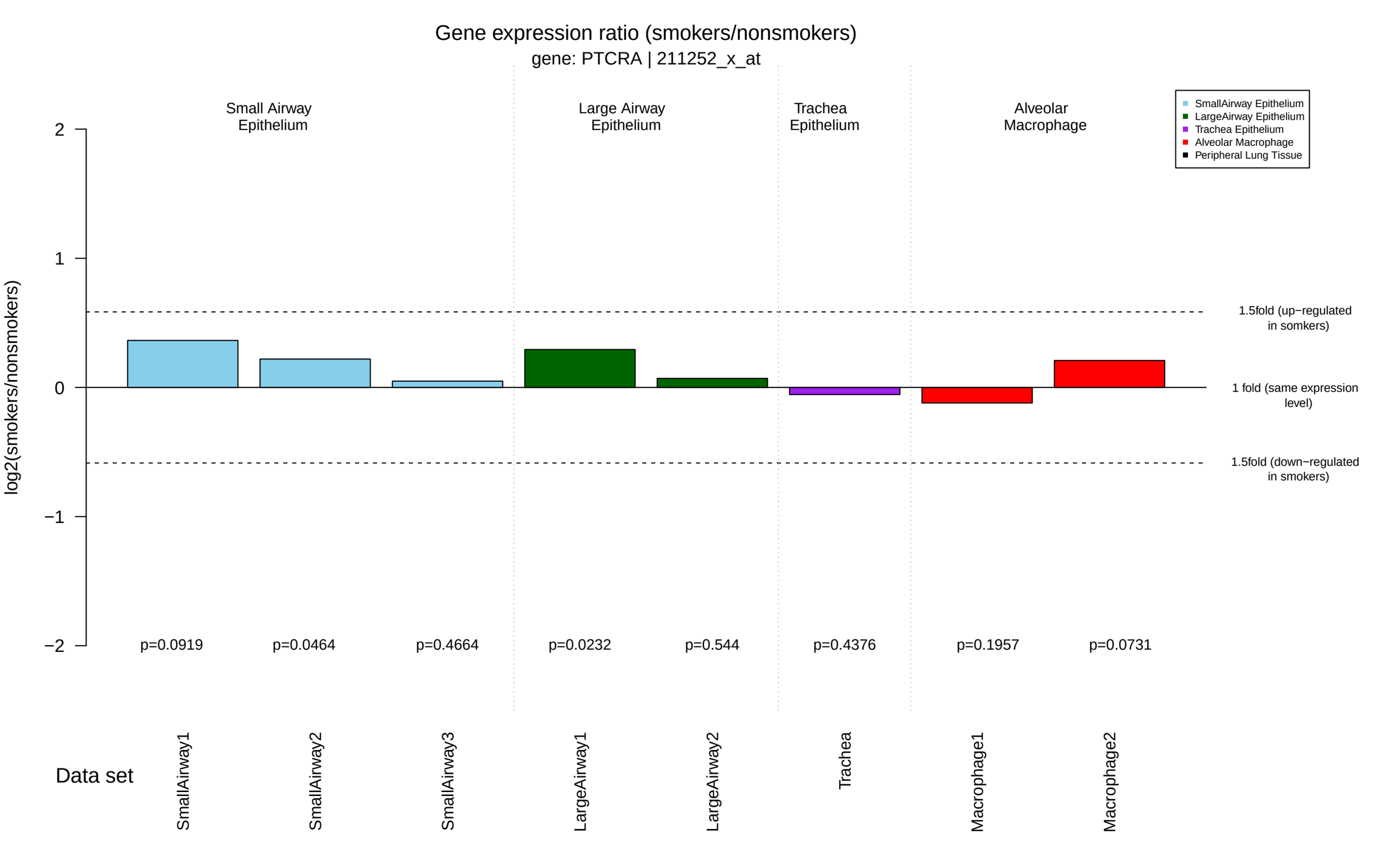
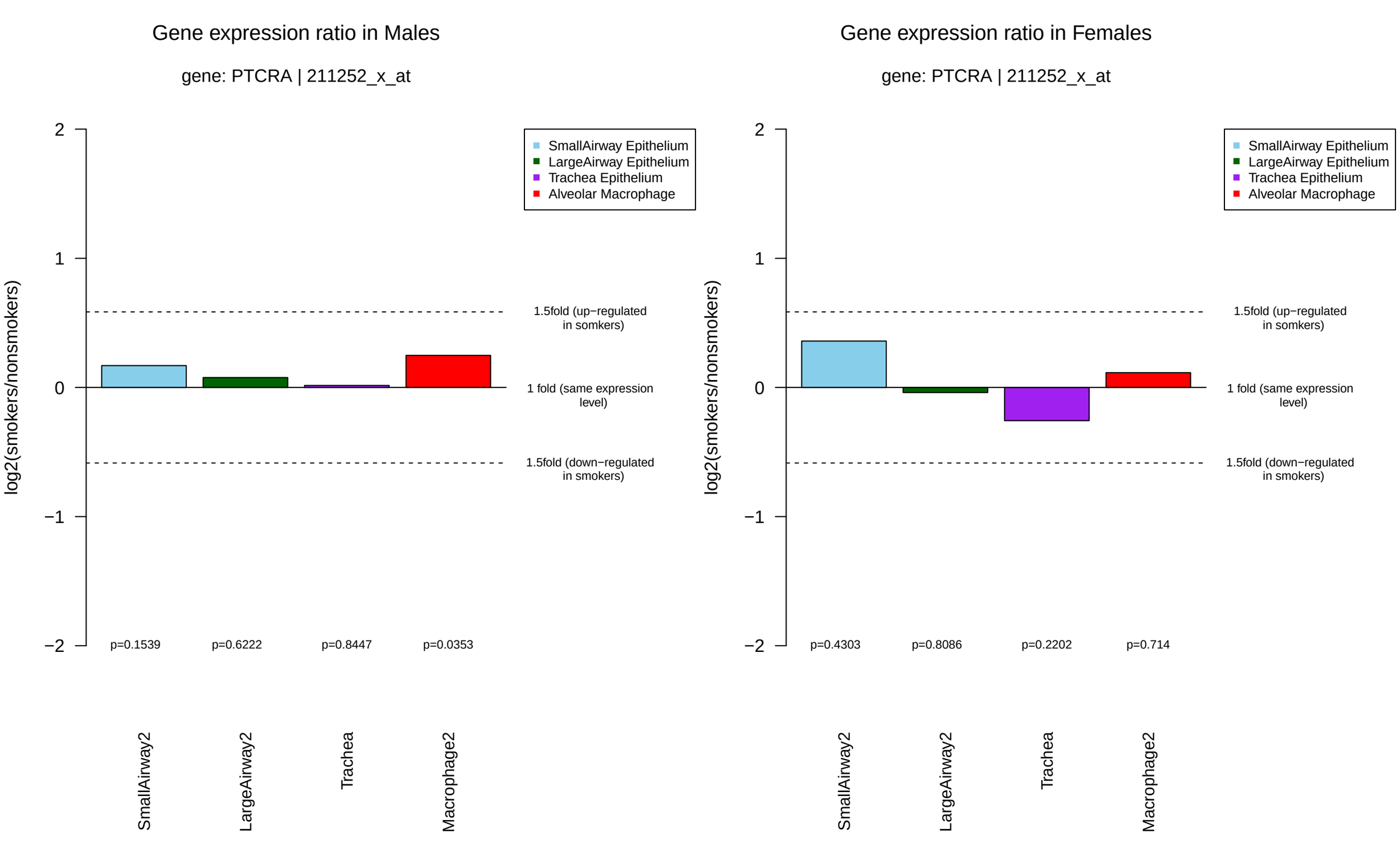
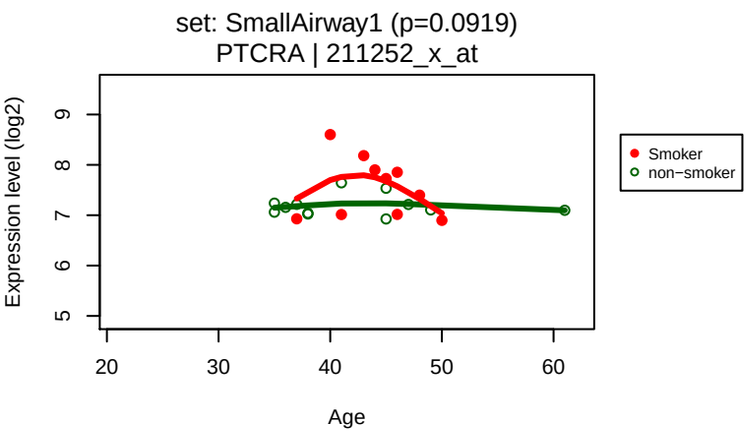
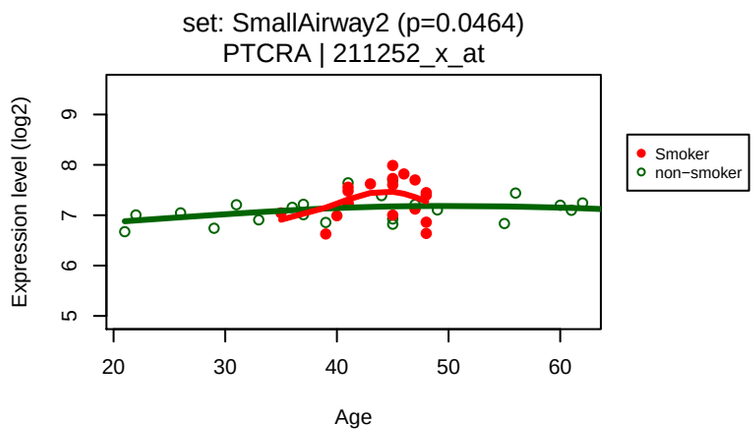
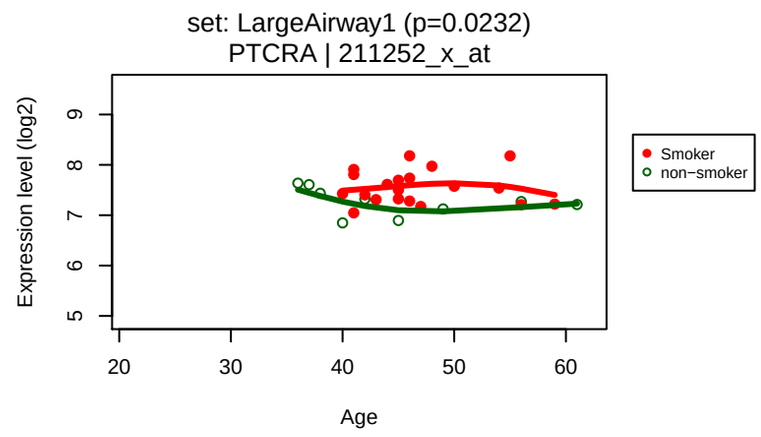

Gene expression of smokers and nonsmokers
gene: PTCRA | 211252_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0919
p=0.0464
p=0.4664
p=0.0232
p=0.544
p=0.4376
p=0.1957
p=0.0731
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

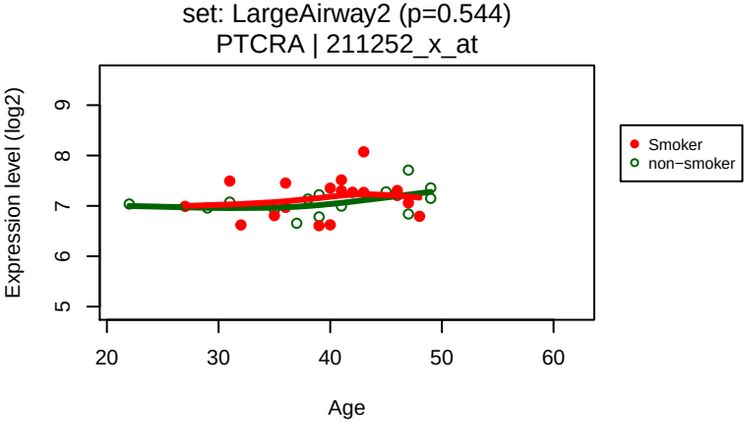

Gene expression ratio (smokers/nonsmokers)
gene: PTCRA | 211252_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0919
p=0.0464
p=0.4664
p=0.0232
p=0.544
p=0.4376
p=0.1957
p=0.0731
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

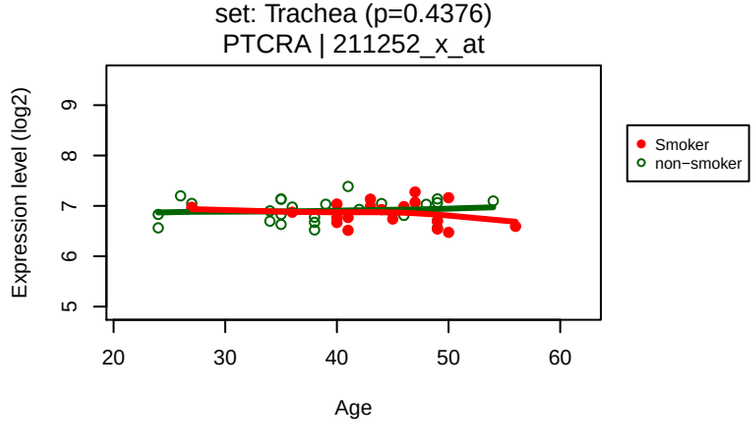

Gene expression ratio in Males
gene: PTCRA | 211252_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1539
p=0.6222
p=0.8447
p=0.0353
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTCRA | 211252_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4303
p=0.8086
p=0.2202
p=0.714
SmallAirway2
LargeAirway2
Trachea
Macrophage2

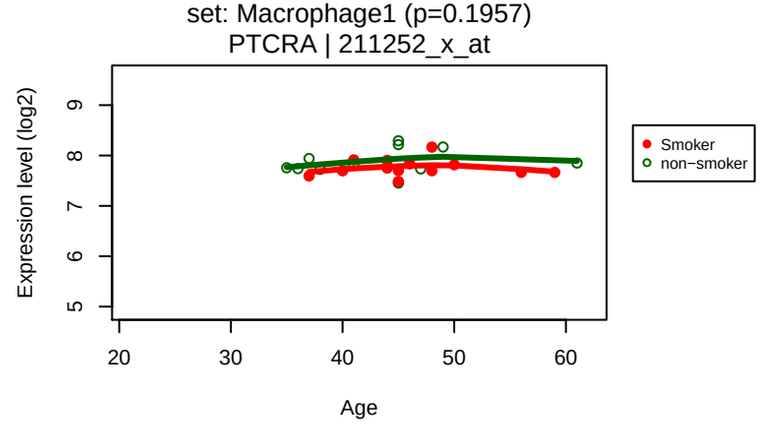

set: SmallAirway1 (p=0.0919)
PTCRA | 211252_x_at
Expression level (log2)
Age
Smoker
non−smoker

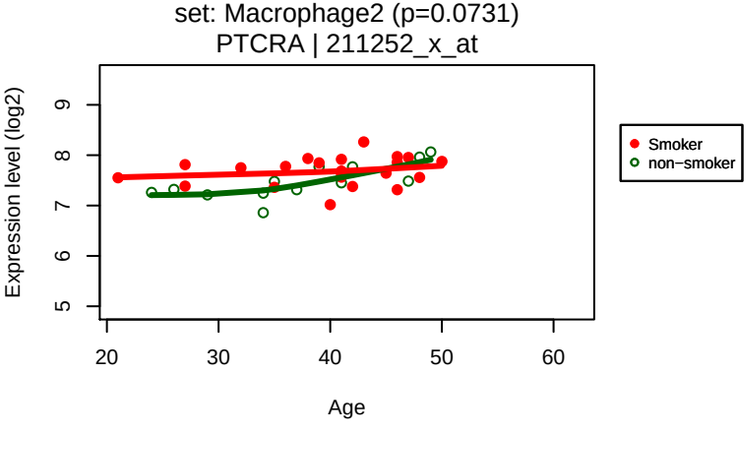

set: SmallAirway2 (p=0.0464)
PTCRA | 211252_x_at
Expression level (log2)
Age
Smoker
non−smoker

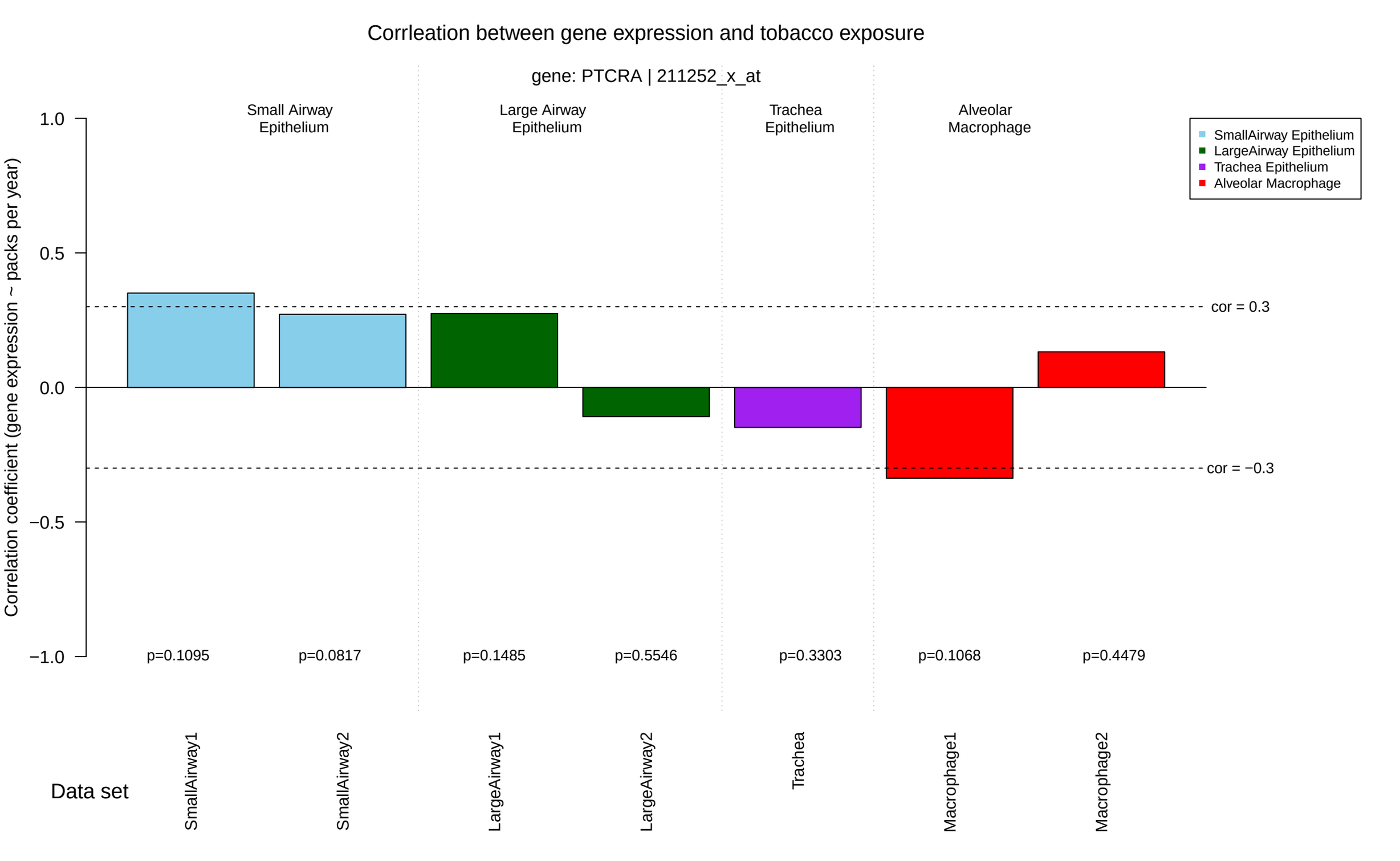

set: LargeAirway1 (p=0.0232)
PTCRA | 211252_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.544)
PTCRA | 211252_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4376)
PTCRA | 211252_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1957)
PTCRA | 211252_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0731)
PTCRA | 211252_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTCRA | 211252_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1095
p=0.0817
p=0.1485
p=0.5546
p=0.3303
p=0.1068
p=0.4479
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2